

ALEXANDRE GOUVEIA RODRIGUES

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EDUCATION

University of Toronto

Sep 2018 – Jun 2023

BASc in Electrical and Computer Engineering · Minor in Engineering Business · Certificate in Artificial Intelligence

PROFESSIONAL EXPERIENCE

Coinbase | Software Engineer

Jan 2025 – Jun 2026

- Architected and built an LLM-powered knowledge platform that consolidated protocol, incident, and infrastructure data from 10+ sources through MCP integrations, REST APIs, scheduled ingestion pipelines, and LangGraph orchestration, giving engineers a single interface for operational investigations.
- Operated production Kubernetes infrastructure for 60+ blockchain protocols, using Docker and Helm to manage node deployments while coordinating upgrades, hard forks, and security patches without disrupting reliability.
- Delivered the end-to-end rollout of new blockchain node infrastructure, from bare-metal server provisioning and Ansible configuration to containerized services and Datadog observability, supporting critical transaction flows.
- Accelerated on-call investigations by building an internal tool that correlated Datadog telemetry, service health, and alerts across distributed systems, reducing mitigation time by approximately 1 hour per incident.

Thomson Reuters | Software Developer Intern

Sep 2021 – Sep 2022

- Automated developer onboarding processes by building RESTful APIs in C# and leveraging AWS Lambda, Amazon Cognito, and Amazon SES, reducing manual operational effort and improving internal developer productivity.
- Developed an internal full-stack authorization testing platform using ASP.NET Core, Angular, and OAuth, enabling developers to validate authentication flows faster and improving reliability of platform integrations.

Amazon.com, Inc. | Software Development Engineer Intern

May 2021 – Aug 2021

- Owned full cycle development of a mobile web feature using React and Redux to help advertisers visualize campaign performance, boosting user engagement and conversion rates by 10%.
- Shipped the customer-facing feature following design reviews and UAT/UI testing; Weblab A/B testing showed widespread adoption with minimal latency, leading to permanent integration into Amazon's website.

SELECTED PROJECTS

Blockchain for Electronic Health Records in Ontario

Sep 2022 – Apr 2023

- Implemented 16 JavaScript chaincode functions and AES-256 encryption logic to support secure user registration, authentication, health-record transactions, and role-based access control within a permissioned blockchain system.
- Built a 14-endpoint Node.js and Express API to bridge the web application and blockchain layer, using JWT-based authorization to enforce role-specific access to chaincode functions and protected health record workflows.

Deep Learning Model for Predicting Song Popularity

Jan 2021 – Apr 2021

- Developed a PyTorch-based Artificial Neural Network (ANN) to predict song popularity from Spotify audio features, achieving 55% test accuracy and outperforming a decision-tree baseline by 10 percentage points.
- Processed, cleaned, and normalized a 174,000-song Kaggle dataset using NumPy, refining audio-feature inputs and class distributions to improve training stability and model performance.

Geographic Information System (GIS) Application

Jan 2020 – Apr 2020

- Created an interactive C++ GIS application with the OpenStreetMap API and EZGL Graphics Library, implementing Dijkstra's and A* algorithms to calculate optimal routes across global city maps.